Electrical power and energy

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Calculate electrical transfer rate and energy use.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 8.*
0.8 A, by charge conservation.

02. Worked example: explain the model
6 V, 2 A, 12 W and 5 s lead to 60 J; quantities are related but not interchangeable.

03. Read and interpret the model
The faster transfer has twice the average power.

04. Apply the physics
12 W; 60 J.

05. Explain your reasoning
10 W for 1 s=10 J; 5 W for 10 s=50 J. Duration matters.

06. Review: connect an earlier idea *Review: Unit 6, Day 17.*
The farther target takes 4 times as long because t=d/v.

07. Exit check
30 J.

08. STRETCH: test the model
6 and 3 s.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 8.*
Parallel 5 Ω; series 20 Ω.

02. Worked example: explain the model
6 V, 2 A, 12 W and 5 s lead to 60 J; quantities are related but not interchangeable.

03. Read and interpret the model
The faster transfer has twice the average power.

04. Apply the physics
9 W; 2160 J.

05. Explain your reasoning
10 W for 1 s=10 J; 5 W for 10 s=50 J. Duration matters.

06. Review: connect an earlier idea *Review: Unit 6, Day 17.*
v≈20 m/s; t≈300 s. Requires the long-wave shallow-water regime and simplified depth; not a damage forecast.

07. Exit check
V=12 V; IV=24 W and $I^2 R$=24 W.

08. STRETCH: test the model
Input 720 J; useful power 10.8 W; efficiency 60%.